Evaluation runs: **eokv** (kv-exactly-once-grove)

generated 2026-11-08 from brain-data 34d8e8fae 2026-11-08 19:23:20 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

No run of the stateless engine on this benchmark.

stateful

No run of the stateful engine on this benchmark.

offline

offline: pair **exactly-once**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	16	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	6	322	conforms	0.6	8	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain05, 2026-11-02
3	20	6072	conforms	0.7	8	10 min, 16 GB	0.1	0.7	0.0	in-memory	brain05, 2026-11-02
4	70	110110	conforms	2.8	22	10 min, 16 GB	0.1	2.7	0.1	in-memory	brain05, 2026-11-02
5	252	1948336	conforms	39.0	228	10 min, 16 GB	0.1	37.6	1.4	in-memory	brain05, 2026-11-02
6	924	33905956	conforms	869.9	2459	10 min, 16 GB	0.2	853.6	16.3	in-memory	brain05, 2026-11-02

offline: pair **no-request-id**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	5	3	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	6	26	20	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	20	125	105	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	70	574	504	violation	0.3	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
5	252	2562	2310	violation	0.5	26	10 min, 16 GB	0.1	0.4	0.1	in-memory	brain05, 2026-11-02
6	924	11220	10296	violation	2.0	126	10 min, 16 GB	0.2	1.2	0.8	in-memory	brain05, 2026-11-02
7	3432	48477		violation	7.5	562	10 min, 16 GB	0.5	2.7	4.8	in-memory	brain05, 2026-11-02
8	12870	207350		violation	33.2	2979	10 min, 16 GB	0.8	7.8	25.4	in-memory	brain05, 2026-11-02
9	48620	880022		violation	215.8	13656	10 min, 16 GB	1.8	32.5	183.2	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): 7 exactly-once bag, 11 no-request-id bag.
skipped: a smaller size had reached its limit: 7 exactly-once bag, 8 exactly-once bag, 9 exactly-once bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **exactly-once**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	16	conforms	4.0	180	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain01, 2026-11-02
2	6	322	conforms	5.0	254	10 min, 16 GB	0.1	5.0	0.0	in-memory	brain01, 2026-11-02
3	20	6072	conforms	45.6	6706	10 min, 16 GB	0.1	45.6	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **no-request-id**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	5	3	violation	3.1	164	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
2	6	26	20	violation	3.4	162	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02
3	20	125	105	violation	4.0	201	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 4 bag exactly-once, 4 bag no-request-id.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **exactly-once**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	16	conforms	2.9	161	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain01, 2026-11-02
2	6	322	conforms	3.1	174	10 min, 16 GB	3.1	0.6	0.0	in-memory	brain01, 2026-11-02
3	20	6072	conforms	3.4	176	10 min, 16 GB	3.2	0.7	0.2	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **no-request-id**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	5	3	violation	2.9	161	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain01, 2026-11-02
2	6	26	20	violation	3.1	174	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
3	20	125	105	violation	3.2	177	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 4 bag exactly-once, 4 bag no-request-id.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **exactly-once**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	16	conforms	2.9	156	1 h, 250 GB		brain01, 2026-11-02
2	6	322	conforms	3.2	170	1 h, 250 GB		brain01, 2026-11-02
3	20	6072	conforms	3.7	172	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-request-id**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	2	5	3	violation	2.9	156	1 h, 250 GB	2	brain01, 2026-11-02
2	6	26	20	violation	3.2	170	1 h, 250 GB	2	brain01, 2026-11-02
3	20	125	105	violation	3.2	172	1 h, 250 GB	2	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **exactly-once**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	16	16	conforms	4.0	180	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-02
2	322	322	conforms	5.1	254	10 min, 16 GB	5.0	0.6	0.2	in-memory	brain01, 2026-11-02
3	6072	6072	conforms	45.7	6706	10 min, 16 GB	45.6	0.7	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **no-request-id**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	5	5	conforms	3.1	164	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	26	26	conforms	3.4	162	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
3	125	125	conforms	4.1	201	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 4 bag exactly-once, 4 bag no-request-id.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **exactly-once**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	16	16	conforms	4.0	180	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain01, 2026-11-02
2	322	322	conforms	5.0	254	10 min, 16 GB	0.6	5.0	0.0	in-memory	brain01, 2026-11-02
3	6072	6072	conforms	45.6	6706	10 min, 16 GB	0.7	45.6	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **no-request-id**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	5	5	conforms	3.1	164	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
2	26	26	conforms	3.4	162	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02
3	125	125	conforms	4.0	201	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 4 bag exactly-once, 4 bag no-request-id.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **no-request-id**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	5	5		conforms*	0.8	159	48 h, 80 GB	0.0	0.8	0.0	merge	pinky13, 2026-11-07
2	26	26		conforms*	0.9	162	48 h, 80 GB	0.1	0.9	0.0	merge	pinky13, 2026-11-07
3	125	125		conforms*	1.6	170	48 h, 80 GB	0.1	1.6	0.0	merge	pinky13, 2026-11-07

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 1 bag exactly-once, 2 bag exactly-once, 3 bag exactly-once.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **exactly-once**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	16		failed			1 h, 250 GB		brain02, 2026-11-03
2	322		failed			1 h, 250 GB		brain02, 2026-11-03
3	6072		failed			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-request-id**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	5		failed			1 h, 250 GB		brain02, 2026-11-03
2	26		failed			1 h, 250 GB		brain02, 2026-11-03
3	125		failed			1 h, 250 GB		brain02, 2026-11-03